

The effect of the source type on wave propagation in a snow layer: 2D numerical comparison

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Abstract

In the following project for the lecture "Research Module in Applied Geophysics", different seismic sources will be set up and wave propagation is modelled using MONDAIC Salvus software. The models set up are a three-layer model (soil, snow, air) and a two-layer model (snow-air), the sources are an acoustic scalar point 2D source, an elastic vector point 2D source as well as a moment tensor source representing a skier on the snow. The different source types show differences in the two model types, which is explained by reflection and transmission of the waves. Furthermore, the waves have very low amplitude velocities in air, which is explained by the material density related to wave velocity. This is useful for the Master Thesis because modelling wave propagation using different sources is snow is required.

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1 Introduction

Snow avalanches are snow masses that descend slopes rapidly (Schweizer et al., 2003). Snow avalanches can occur as different types, such as snow slab avalanches, where a slab of snow slides (Gaume et al., 2017). Such avalanches are among the most prominent hazards in mountainous regions (Gaume et al., 2017), they can be triggered by rapid loading near the surface, for example by a person skiing over them, or by a dynamite source (Schweizer et al., 2003). The cracks caused by this propagate and eventually lead to material failure, the release of the slab is caused by an increase of shear stress in a weak layer in the snowpack (Habermann et al., 2008).

Understanding the signatures of different types of sources is crucial for understanding crack propagation. This research project aims to compare simulated wave propagation of three different source types (pure acoustic pressure source, elastic vector source, and a moment tensor source in 2d) in two different models, which are earth-snow-air and snow-air. The aim of this comparison is to determine which models and source types are useful to understand crack propagation in snow leading to slab avalanches, which will be analyzed in the Master Thesis. These considerations are not only based on the results of the different types of simulations, but also computation time and memory.

2 Methodology

2.1 Introduction to Mondaic Salvus Software

Salvus is a numerical modelling software by Mondaic (“Get Salvus”, n.d.). It simulates wave propagation in different media using functions predefined by the software. Salvus has a large database of predefined seismic source functions, where the source parameters such as forces in different directions can be changed. These sources are placed in a medium defined by the user, an unstructured mesh is created. These meshes are created based on points obtained by finite-element methods, it uses a set number of Gauss-Lobatto-Legendre points for every element (“Get Salvus”, n.d.). In this case, a simulation is run which computes particle velocity at each of these mesh points at a timestep interval given.

2.2 Spectral element wave propagation with Salvus - acoustic & elastic case

The first simulation which was set up is set in a homogenous domain 1m by 1m. The density (ρ) as well as p- and s-wave velocities have values of snow, which were taken from Guillemot et al., 2021, table 2, the values for soil and air were taken from “Seismic Velocities of Rocks and Various Materials — GPG 0.0.1 Documentation”, n.d. All the medium parameters which were used in the three different models can be seen in table 1. The source, which is located halfway at the top of the domain ($x=0.5$, $y=1$) is a scalar point 2D source with force in negative y-direction. The field which was recorded in this case is $\frac{\partial \phi}{\partial t}$, meaning the particle velocity. This simulation was created to verify the setup of the domain and get more familiar with simulation setup. The s-waves were not recorded in this simulation, this is to minimize simulation time. This simulation will not be analyzed as a result.

Table 1. Material Parameters (density and p and s-wave velocities used in the models)

Material	ρ [kg/ m^3]	Vp [m/s]	Vs [m/s]
Snow	180	300	150
Air	1.2250	332	0
Soil	2000	2200	880

Next, a simulation with a snow-layer of 1m thickness and an 1m thick air layer was created, both layers have an x-extent of 30m. This was done by creating numpy-arrays with the edges of the layers in x and the y direction, so for the snow layer this would be [0,30] in x and [0,1] in y. Then, the model parameters were also defined as arrays. Then, a mesh was generated by interpolation between the edge points and the material parameters were assigned to the appropriate layers. Furthermore, absorbing boundaries were added to the bottom of the snow layer, the top of the air layer and all sides. This was done so that there are no reflections from the model boundaries, which is closer to a real-life scenario. First, a pure acoustic simulation with a scalar point 2D source was run, the source was located halfway in the snow layer, the simulation recorded the acoustic velocity field. The parameters used for simulation can be seen in table 2. Next, the simulation was run for the elastic case with two different source types: a vector point source with $f_y=-1$ and a moment tensor source with $m_{xy}=3e4$, the m_{yy} and m_{xx} components were chosen to be 0. The vector source represents an explosive source in the snow, whereas the moment tensor source represents a slip on a surface. The m_{xy} component of the force was chosen assuming a 70kg skier on the snow. The normal force was calculated according to $F = m * g = 70 * 9.81 = 700N$, the frictional shear force between the skis and the snow was assumed to be 100N, the slip distance 0.1m and the slip area on two skis $0.3 m^3$, the shear modulus μ of snow 10^6 Pa. This leads to $M_0 = \mu * A * D = 10^6 * 0.3 * 0.1 = 3 * 10^4 N$.

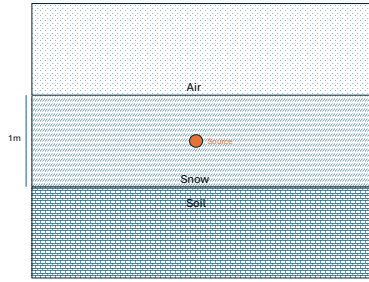


Fig. 1. Layered model setup. Each of the layers has a 1m extent in y, and 30m extent in x. Source location indicated by orange dot.

Lastly, a three layered model was set up, which consists of a 1m soil layer, a 1m snow layer and an 1m air layer, all with 30m lateral extent, was set up as described above and shown in figure 1. Again, absorbing boundaries were added, instead of the bottom of the snow layer the absorbing boundary conditions were added to the bottom of the soil layer. The medium properties are shown in table 1. Again, first the pure acoustic case was considered, with the same source type and field as described above. Then, the elastic case was recorded with a sources of the same types as above located halfway in the snow layer. The simulation parameters used can be seen in table 2.

Table 2. Simulation parameters used

Parameter Name	Value
Source Center Frequency	700Hz
Event End Time	2s
Sampling Interval	200

2.3 Numerical Considerations for wave physics

To set up the simulation parameters in table 2 and the different models, a few physical considerations were needed.

Firstly, the extent of the layers needs to be plausible. Not only do avalanches usually occur over large spatial extent, the lateral extension of the model also needs to be large enough so that far-field effects of the waves can be observed. A function was written in python which performs these calculations. This was determined by taking the maximum and minimum speed present in the model, the speeds are the same as in table 1. The source wavelet, which was a ricker wavelet in all cases, was fast-fourier transformed (fft). From the frequency values of the wavelet, the minimum and maximum frequency were determined. The smallest and largest wavelengths to be resolved were determined according to $\lambda_{min} = \frac{v_{min}}{f_{max}}$ and $\lambda_{max} = \frac{v_{max}}{f_{min}}$, where λ is the minimum or maximum wavelength, v is the velocity and f is the frequency determined by the fft. To observe far-field effects, the medium needs to be much larger in the x-direction than the maximum wavelength to be resolved. In this case, the maximum wavelength determined was about 2cm, which means that an x-extent of 30m is sufficient. The vertical thickness of the layers was determined by real-world considerations, meaning that they were determined by what is plausible, a 40m snow layer for example would not be very likely.

The center frequency of the source was determined by the layer thicknesses. The minimum thickness of a layer that can be resolved is about $\frac{\lambda}{4}$, the wavelength can be converted to frequency according to $f = \frac{v}{\lambda}$ ("Seismic Resolution", 2018). The velocity taken for this consideration was the maximum velocity present in the model. Based on the layer thicknesses and speeds present, a center frequency of 700Hz was chosen to make sure all layers are resolved well. This is a very high center frequency, the DAS data and modelling used in the thesis will use a much lower center frequency.

The simulation sampling interval is a tradeoff between simulation time, memory needed and resolution. The larger the sampling interval, the fewer samples in time. The appropriate time step was determined by the Nyquist sampling criterion where there need to be at least two sampling points per period: $f_{sampling} = \frac{1}{frequency} * \frac{1}{2}$, where the frequency is the maximum meaningful frequency of the wavelet, which can be obtained by plotting the wavelet.

3 Results

3.1 Pure Acoustic

As mentioned in the methods section, the source used in the pure acoustic case is a 2D scalar point source in negative y-direction. Figure 2 shows the pure acoustic case of the snow-air model. As shown in 2, the acoustic waves travel in the snow layer only. Figure 3 shows the pure acoustic case of the earth-snow-air model, where earth extends from 3m to 2m, snow from 2m to 1m depth and air from 1m to 0m, here, the acoustic waves from the scalar point source also only travel in the snow and earth layers. Figure 2 shows wavefronts over several meters, whereas figure 3 shows much smaller wavefronts. At timestep 3 in the snow-air model, shown in figure 2a, the wavefield has already propagated, while at the same same timestep for the earth-snow-air model, shown in figure 3a, the propagation is much less far.

Slower propagation of the acoustic wavefield for the three-layer model can also be seen in 3c and 3d, it takes the wavefield 0.05273 seconds to reach the end of the model domain, while in the two-layer case the wavefield has almost reached the edge of the domain at 0.02551s, shown in 2b.

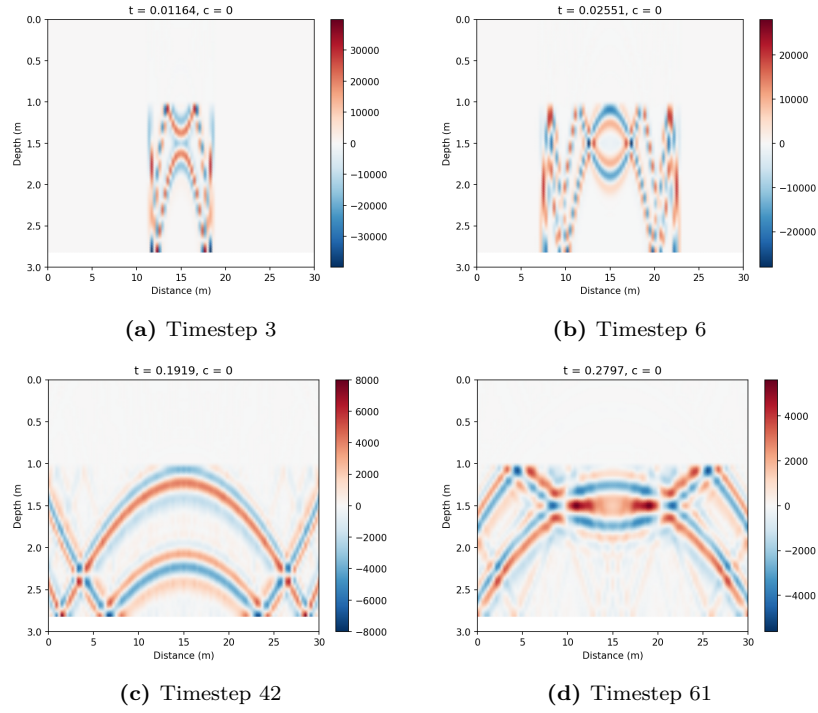


Fig. 2. Pure Acoustic snow-air model velocity outputs at selected timesteps.

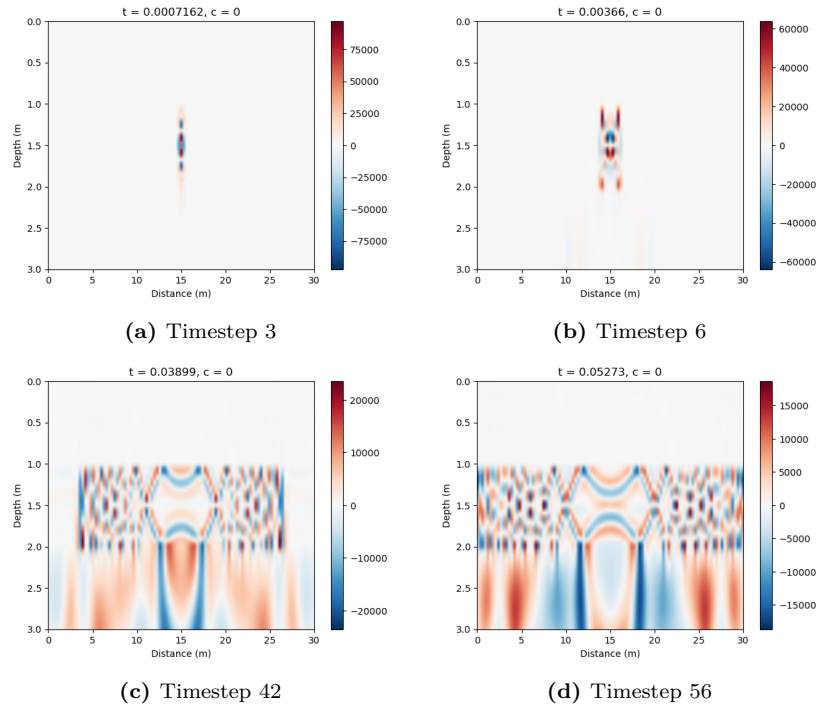


Fig. 3. Pure Acoustic layered model velocity outputs at selected timesteps.

The two layer model shows much less reflection at the snow-air interface, compared to the three layer model, for example in figure 3d. It is also important to note that the layers are a bit less thick in the 3 layer case compared to the 2 layer case.

3.2 Elastic

The observations made about the speed of the wave propagation are also true for the elastic case with a vector source, the vector source has a negative y-direction. The two-layer case is shown in figure 4, the three layer

case in 5. The waves from the vector point 2D source also do not propagate through the air in both cases, in the three layer case in figure 5 also does not show propagation through the earth layer.

3.2.1 Vector Source

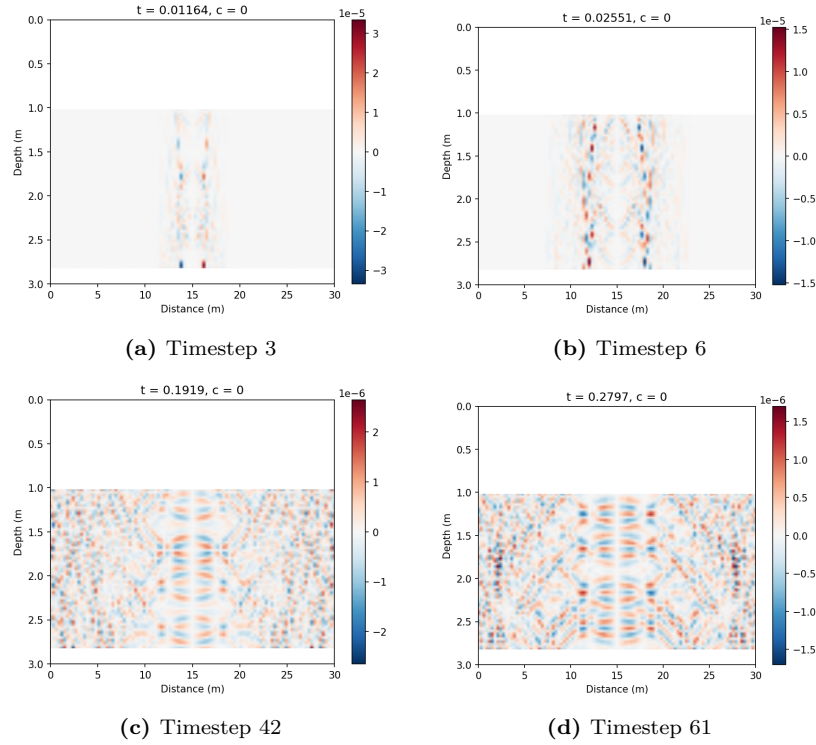


Fig. 4. Elastic snow-air model outputs with 2D vector source in negative y at selected timesteps.

While the wavefields at larger distances look similar for both cases, the two layer case shows a behavior different from the three layer case at later timesteps in 4c and 4d close to the source location. Here, ellipsoidal patterns can be seen, these are located at the same horizontal distance as the source location in all y-positions. Such patterns are absent in the three layer case.

While the two layer case shows distinct patterns close to the source, the three layer case in figure 5 shows wavefronts of more extreme velocities propagating at later timesteps in 5c and 5d. In the two layer case, particle velocities show no clear areas with extremes, velocities have the same magnitude everywhere. In addition, while the two layer case in 4c and 4d shows patterns at the same x-location as the source, the same timesteps in the three-layer case shown in figures 5c and 5d show a blindness at the x-location of the source, there is no velocity field above and below the source.

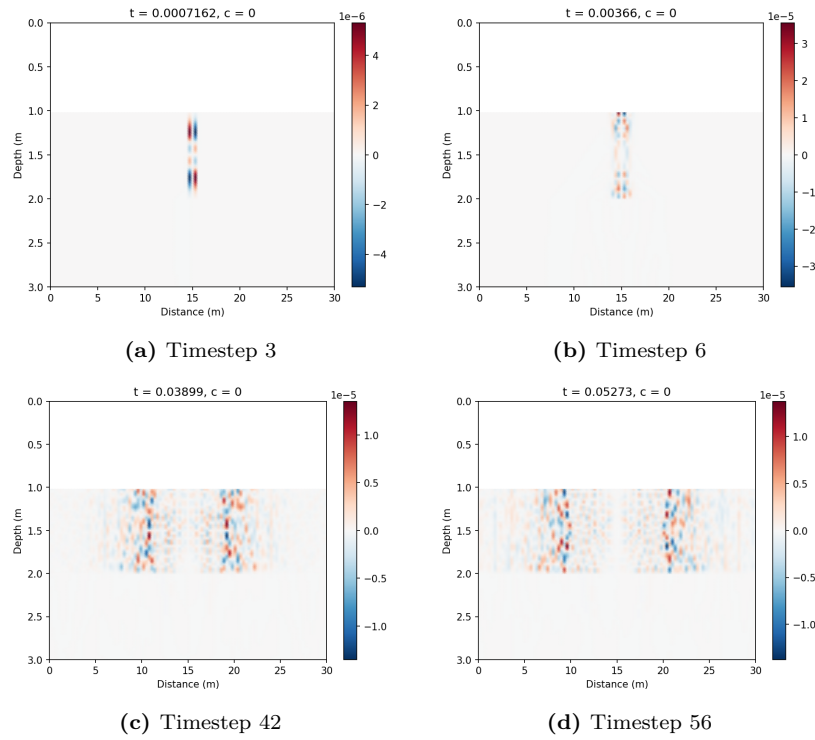


Fig. 5. Elastic layered model outputs with 2D vector source in negative y at selected timesteps.

3.2.2 Moment Tensor Source

The moment tensor source representing a skier sliding on the snow also does not show any wave propagation through the air layer for both the two layer case in figure 6 and three layer in 7. The source is located within the snowpack because as mentioned in the introduction, avalanches are usually triggered by a weak layer failing in the snowpack. Here, the difference in distance of the wavefield propagation between the two and three layer case is not as remarkable as the other source types, at timestep 6, figures 6a and 7a show similar propagation distances in the horizontal direction.

The wavefield in the three layer case shows complex patterns in the snow layer at later timesteps in 7c and 7d, like for a 2D vector source, the area above the source does not show a wavefield. In the three layer case, the same timesteps show much simpler patterns in the earth layer, so below 2m depth. The wave propagation pattern in the two layer case at the same timesteps in figures 6c and 6d shows much simpler wavefronts similar to those generated in the pure acoustic case.

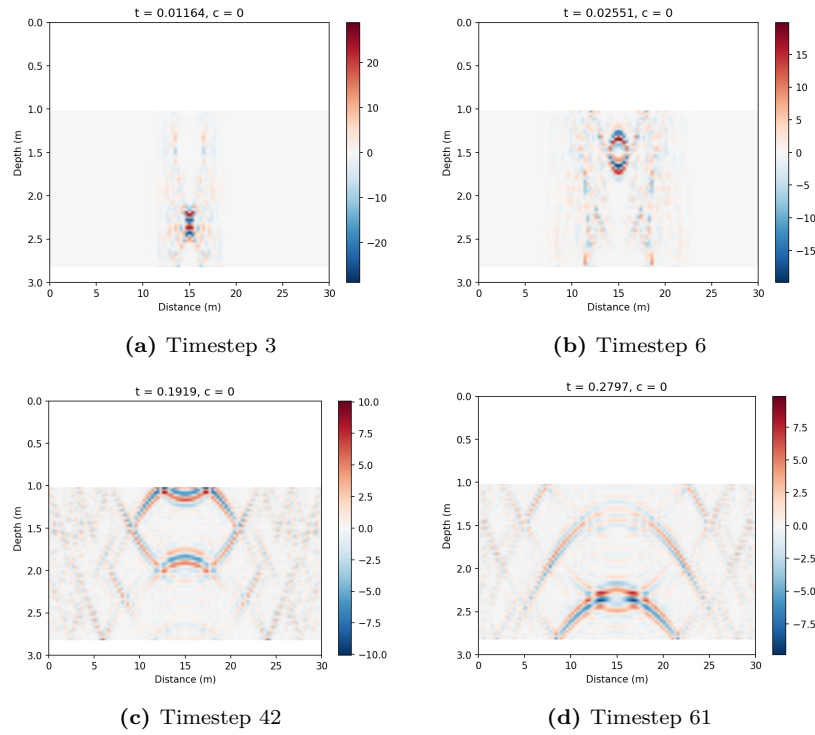


Fig. 6. Elastic snow-air model outputs with 2D moment tensor source with $m_{yy}, m_{xx}=0$, $m_{xy}=3e4$ at selected timesteps.

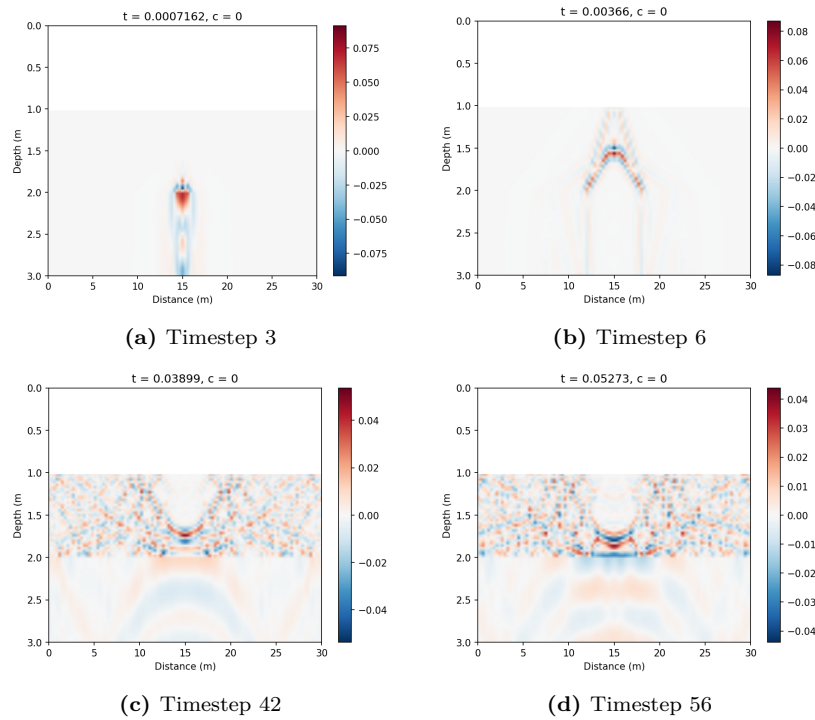


Fig. 7. Elastic layered model outputs with 2D moment tensor source with $m_{yy}, m_{xx}=0$, $m_{xy}=3e4$ at selected timesteps.

3.3 Analysis of wave propagation in air

As the above results have shown, propagation of waves can not be seen in any of the results. This is not the case when a source is being placed in the air, as shown in figure for the pure acoustic case in the snow air model. The wave propagation in snow looks similar to that in figure 2a, which is the same timestep and source type, with the source placed in the snow.

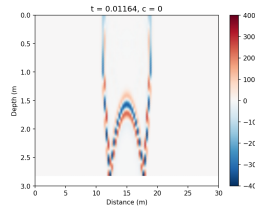


Fig. 8. Pure acoustic case in snow-air model with source placed in the air.

Alternatively, wave propagation can also be plotted in air only, with the source still located in snow. This is shown in figure 9 for the two-layer pure acoustic case. When comparing the pressure wavefield in the air in figure 9 to that in snow in figure 2, the wavefields in the air seem to be a continuation of those in snow. For the pure acoustic case, there are clear wavefields in the air when plotting them separately, but in the case of the velocity field, the plots show no significant wavefields.

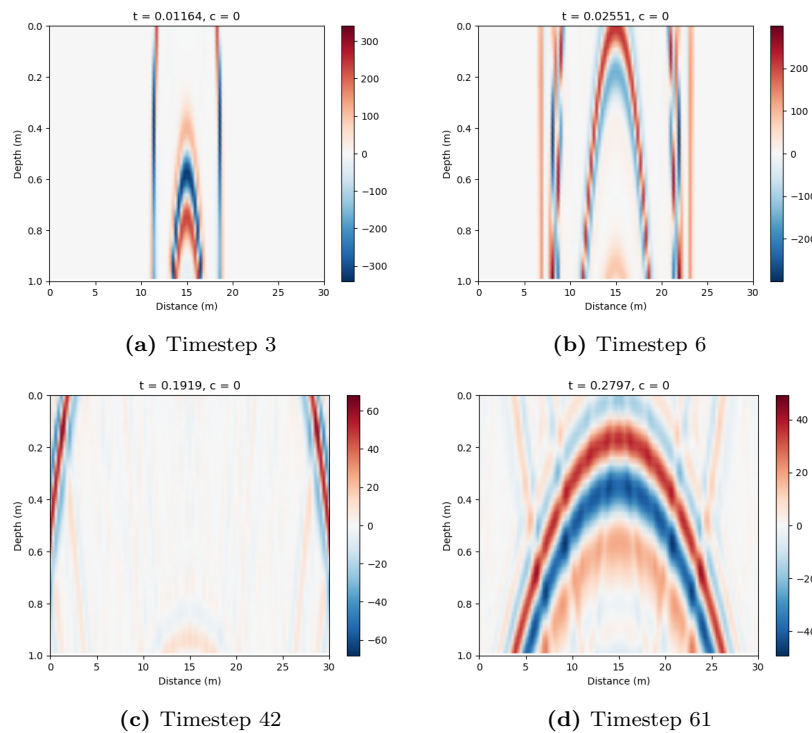


Fig. 9. Elastic layered model outputs with 2D moment tensor source at selected timesteps.

4 Discussion & Outlook

4.1 Effects observed at the snow-air interface

As observed in the pure acoustic results in section 3.1 as well as in the elastic case discussed in 3.2.1 and 3.2.2, the final plots show no wave propagation in the air layer, unless one plots the pressure wave in the air layer only.

As seen from the medium parameters in table 1, the s-wave velocity in air is 0, because it is a fluid. This means that at the interface, all incident s-waves are either reflected into the medium below or transformed to p-waves at the boundary (Borcherdt, 1982). This explains why only the pressure field can be seen in the plots in figure 9. The contrast between complexity of the wavefields in soil and snow shown for example in figure 7c can also be explained by this. The snow layer has much more reflections of s-waves from the snow-air boundary, whereas s-waves can be refracted at the snow-soil boundary.

Furthermore, the medium properties in table 1 also show a strong contrast in density ρ between all three mediums. According to the formulations of wave speed described by Aki and Richards, 2002 and Denneman et al., 2002, where 1a describes the speed of a p-wave in a pure acoustic medium, 1b the p-wave speed with Lamé-parameters λ and μ and 1c describes the s-wave speed.

$$c = \sqrt{\frac{K}{\rho}} \quad (1a)$$

$$V_p = \sqrt{\frac{\lambda + 2\mu}{\rho}} \quad (1b)$$

$$V_s = \sqrt{\frac{\mu}{\rho}} \quad (1c)$$

All of these equations show a dependance of wave speed to density, which means that low density leads to high velocities. When looking at the bulk modulus K of a material, which describes how resistant a material is to change (“Bulk Modulus”, 2016).

$$K = \rho \left(V_p^2 - \frac{4}{3} V_s^2 \right) \quad (2)$$

which is dependent on the density and velocity of a material and displacement, the air layer with low densities has a high speed and a low K compared to the other layers. This means that there is a very large impedance contrast, which causes most of the waves to be reflected into the snow layer, which leads to small amplitudes in the air layer.

4.2 Differences between two and three layer model

The faster propagation of the wavefield in the two-layer model compared to the three-layer model observed for example in figures 2 and 3, even though the time steps have the same number, can be explained by the fact that the timesteps are much smaller in the three-layer case. As shown in the simulation parameters in table 2, both cases have the same end time and sampling interval in time. This can be explained by the equations 1. Since the density as well as wavespeeds in the soil layer are much higher than in the snow layer, the waves which are transmitted into the lower layer, so the soil layer, travel much faster compared to the ones in the snow layer (Treitel & Robinson, 1966). Because of the Courant-Friedrichs-Lewy stability condition, which is used by Salvus, which states that in two dimensions

$$C = \frac{u_x \Delta t}{\Delta x} + \frac{u_y \Delta t}{\Delta y} \leq C_{\max} \quad (3)$$

Where C is the courant number, u the velocity in x or in y direction, Δx the step in x -direction, Δy the step in y and Δt the timestep (Courant et al., 1967). Because the velocity in the bottom layer in the soil is much higher, this means that in the two layer case, the timestep chosen for stability can be larger because the velocities are lower.

The difference in wavefront patterns between the snow-air and the soil-snow-air model can be explained by transmissions and reflections from the different surfaces. Because the three layer case has one interface more at which the waves are reflected and refracted, it makes sense that the wave propagation patterns observed in the snow layer are more complex in the three-layer case. In other words, in the three layer case there are reflections and refractions from the snow-air boundary as well as the snow-soil boundary, whereas in the two layer case, those only happen at the snow-air boundary.

4.3 Comparison of different source types

The results show significant differences in propagation patterns between the different source types. To understand the patterns shown, figure 10 shows the directions of the force of the source with an orange arrow. Eaton, 2009, was used to help visualize the moment tensor source direction.

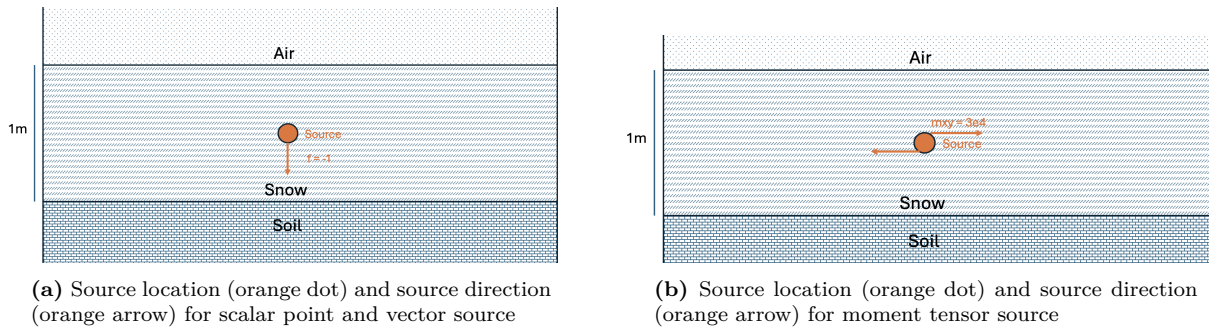


Fig. 10. Source types used with directions of source force indicated by orange arrow. The components not shown are zero.

When comparing the scalar point 2D source and the vector point 2D source in figures 2, 3, 4 and 5 to the source direction in 10a, it becomes clear that the particle velocity patterns are shaped like a half-ellipse perpendicular to the direction of propagation and opposite to the direction of propagation. This makes sense because there is expansion downward, causing the ellipsoidal field, which is mirrored. In the moment tensor source case, whose directions are shown in figure 10b and the corresponding results in 6 and 7, both cases show half-ellipsoidal particle velocity propagation in negative y-direction. The motion caused by the moment tensor source is shown in the beach ball in figure 11 taken from “Moment Tensors - a Practical Guide | mXrap”, 2025, where the blue arrows indicate direction of motion by the source. This means that there is expansion in the horizontal direction, indicated by the black area in 11, which corresponds well with the velocity fields observed, since the tangential motion between the expansion and compression indicated by the purple arrows has the same shape as the wavefronts observed.

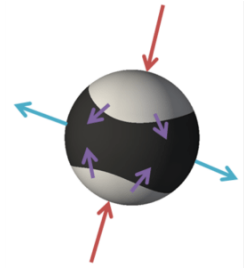


Fig. 11. Beach ball for moment tensor focal mechanism.

4.4 Outlook to Master Thesis Project

This research project serves as an useful preparation for the following master thesis. This project was crucial in learning how to use Mondaic Salvus modelling software, which will also be used in the thesis. Furthermore, the models generated and the results from different source types serve as a basis for the thesis project. From this preparation, I now understand for example that, as shown by the results of the different models and source types, including the air layer is not strictly necessary because there is no significant wave propagation in that layer, however the number of layers is important for complexity of wavefields caused by reflection. This will help to optimize model computation time in the thesis. The models used in this research project are much simpler than the ones which will be used in the master thesis, for example there is no topography included in these models and the snow layer is homogenous, which means that new models will need to be made for the thesis in order to represent the conditions on the mountain more accurately.

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6 Data Management

The following table 3 describes the data generated for the project and where it is stored:

Table 3. Description of data types and storage locations

Data type	Description	Format	Storage
Jupyter Notebooks	Jupyter notebooks containing simulation, model setup and plotting code	.ipynb	GitHub repository
Volume output data	Waveform data output by Salvus simulations	.h5 and .xdmf	Local (Harddisk)
Figures and animations	Wavefield output plots and animations	.jpeg and .mov	Local and polybox
Report	Latex document containing report	.tex	Github repository

A new data management plan will be needed for the master thesis, as the simulation files generated are very large and cannot be stored locally. The external storage of simulation files is also important for reproducibility.

7 Appendix

The animations for the wavefield propagation of each source type can be found on polybox: <https://polybox.ethz.ch/index.php/s/97aa3x2o2FtnMS5>

The full code for all simulations run is stored on the github repository: https://github.com/s-bachmann/researchmodule_agp

7.1 Declaration of Originality

All the text, figures and ideas written in this report are my own, unless cited otherwise. Chatgpt was used to help with L^AT_EX formatting and debugging of L^AT_EX output. Copilot was used to attempt to optimize code written by me, none of this code made it to the final version.